Edge Detection and Feature Extraction

DSP Lab Course WS 2025/26

Authors:

Manikandan Ramesh (manikandan.ramesh@fau.de), Andreas Löw (andreas.loew@fau.de)

Supervisor: Alexander Kopte

Introduction

Edge detection acts as a vital preprocessing stage in image analysis, converting raw pixel data into meaningful structural contours while filtering out redundant information. This project examines a suite of classical and advanced detection techniques, evaluating the performance of first-order gradient operators like Roberts, Prewitt, and Sobel alongside the noise-resilient Laplacian of Gaussian (LoG) method.

A core component of this work involves the Canny edge detection pipeline, which integrates Gaussian blurring, gradient magnitude analysis, non-maximum suppression, and hysteresis thresholding to produce precise, single-pixel edge maps. Moving beyond general boundary detection, the project emphasizes feature extraction by identifying junctions and corners in quadratic checkerboard patterns, assessing the robustness of these algorithms for critical computer vision tasks such as 3D reconstruction and camera calibration.

Algorithm

The implementation of edge detection in this project follows a structured mathematical approach, transitioning from simple first-order derivative operators to the sophisticated multi-stage Canny pipeline. Each method aims to identify intensity discontinuities within the image, which signify physical boundaries or features.

Gradient-Based Operators (Roberts, Prewitt, and Sobel)

These algorithms utilize convolution with discrete kernels to approximate the first-order derivative of the image intensity function. The process follows these steps: Convolution: The input image I is convolved with horizontal (G_x) and vertical (G_y) masks. Magnitude Calculation: The gradient magnitude $M(x, y)$ representing the edge strength, is calculated as:

$$M(x, y) = \sqrt{G_x^2 + G_y^2} \tag{1}$$

Phase Calculation: The gradient direction is determined by:

$$\theta = \arctan\left(\frac{G_x}{G_y}\right) \tag{2}$$

Laplacian of Gaussian

The LoG operator is a second-order derivative method designed to find zero-crossings. Because second derivatives are extremely sensitive to noise, the algorithm incorporates a Gaussian smoothing step:

1. Smoothing: The image is convolved with a Gaussian kernel $G(x, y)$ to remove high-frequency noise.

2. Laplacian Operation: The Laplacian operator ∇^2 is applied to the smoothed image:

$$\nabla^2 f = \frac{\delta^2 f}{\delta x^2} + \frac{\delta^2 f}{\delta y^2} \tag{3}$$

The combined operation is defined as:

$$\log(x, y) = \frac{-1}{\pi\sigma^4}\left(1 - \frac{x^2 + y^2}{2\sigma^2}\right) e^{\frac{-x^2+y^2}{2\sigma^2}} \tag{4}$$

The Canny Edge Detection Algorithm

The Canny detector is an optimized multi-stage algorithm designed for thin, accurate, and noise-resilient edge maps. The implementation follows the criteria proposed by John Canny:

1. Noise Reduction: Gaussian filtering is applied to the image:

$$f_{smooth}(x, y) = I(x, y) * G(x, y, \sigma) \tag{5}$$

2. Gradient Analysis: Sobel kernels are applied to find the edge magnitude and orientation.

3. Non-Maximum Suppression (NMS): This thins the edges by suppressing pixels that are not local maxima in the direction of the gradient.

4. Double Thresholding: Pixels are classified into three categories:

   a) Strong: $M(x, y) \geq T_{high}$

b) Weak: $T_{low} < M(x, y) \leq T_{high}$

c) Suppressed $M(x, y) \leq T_{low}$

5. Hysteresis Tracking: Weak edges are only included in the final output if they are connected to a strong edge.

Implementation

The project was realized in a Python-Jupyternotebook for good read- and traceability of the code and the resulting graphs. For the algorithms the libraries NumPy, SciPy and SKImage were used and for the visualization MatPlotLib.

Gradient-based edge detectors were implemented using convolution with Roberts, Prewitt, and Sobel kernels to get the horizontal and vertical derivatives, which were than combined to the gradient magnitude using Equation 1.

For spectral analysis, a two-dimensional Fourier transform was applied. Convolution kernels and images were zero-padded to achieve a higher frequency resolution. The resulting spectra were visualized using logarithmic scaling and three-dimensional surface plots.

Noise robustness was analyzed by applying deterministic cosine noise and additive Gaussian noise to the rotated checkerboard images before the edge detection. Using the spectral analysis the noise was than visualized.

Second-order edge detection was implemented using different approximations of the Laplacian operators and zero-crossing detection. The Marr-Hildreth approach was realized by applying Gaussian smoothing before the Laplacian.

The Canny detector was implemented according to the document [1] using gaussian filtering, gradient-based edge detection, non-maximum suppression, double thresholding, and hysteresis-based edge tracking.

Evaluation Results

The evaluation was performed on synthetic checkerboard images and randomly selected BSDS500 images. The checkerboard experiments allow controlled analysis of edge orientation, while natural images provide qualitative assessment under realistic conditions.

Figure 1 shows the responses of the gradient-based detectors on a rotated checkerboard image. Roberts produces thinner edges than Prewitt and Sobel. The Sobel edge detection is slightly smoother compared to Prewitt.

The frequency-domain visualization in Figure 2 confirms that all gradient operators

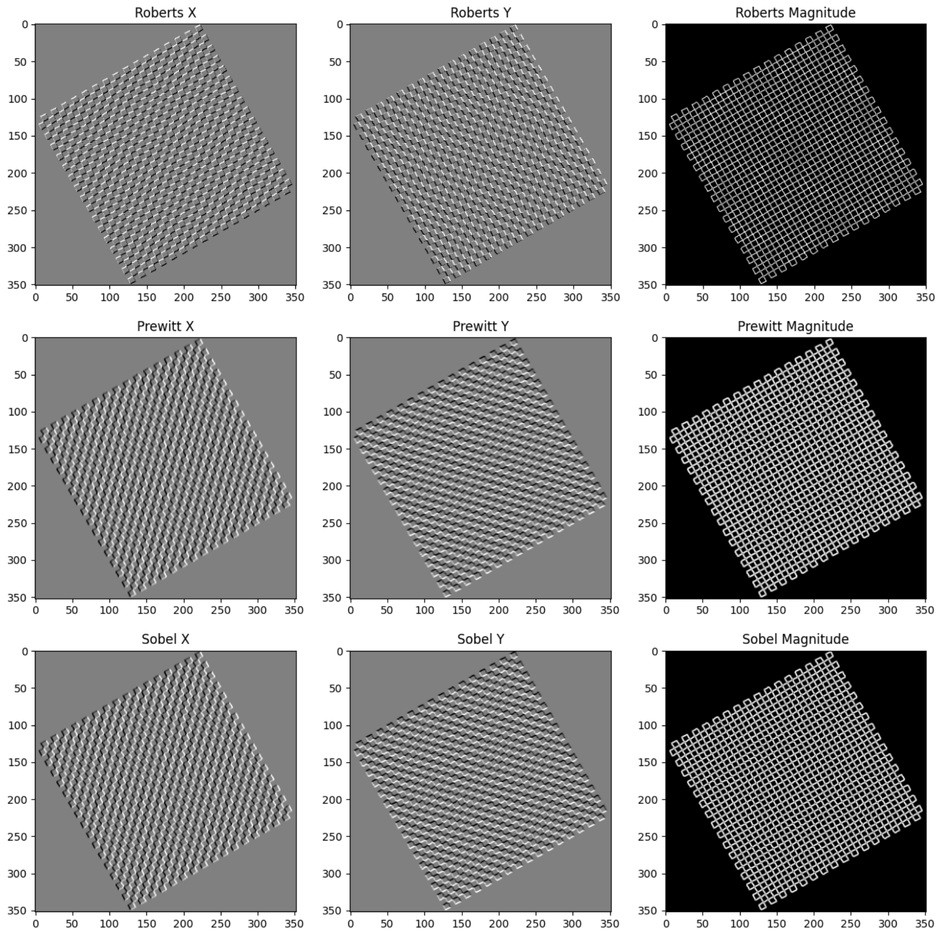

Figure 1: Gradients-based edge detection

act as directional high-pass filters. The strongest spectral response appears perpendicular to detectable edge orientations, explaining the observed directional selectivity.

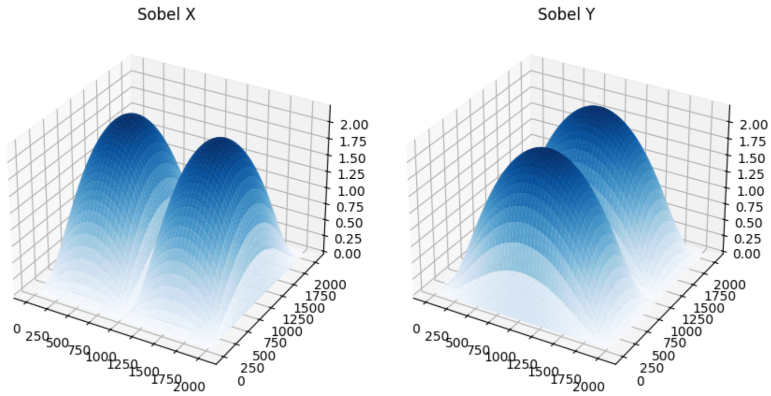

Figure 2: Spectral analysis of the Sobel kernel

The influence of noise is illustrated in Figure 3. The graphs in the upper row show the frequency responses of the derivative of the rotated checkerboard with without and with cosine or gaussian noise, in the lower row the magnitude of the edge-detection. The cosine noise introduces new peaks at its respective frequency while not influencing other frequencies. Whereas the gaussian noise reduces the variance in the checkerboard's spectrum. This can be seen smoother surface the derivative's spectrum and the better visibility of the peak of the magnitude's spectrum, which stems from checkerboard's frequency.

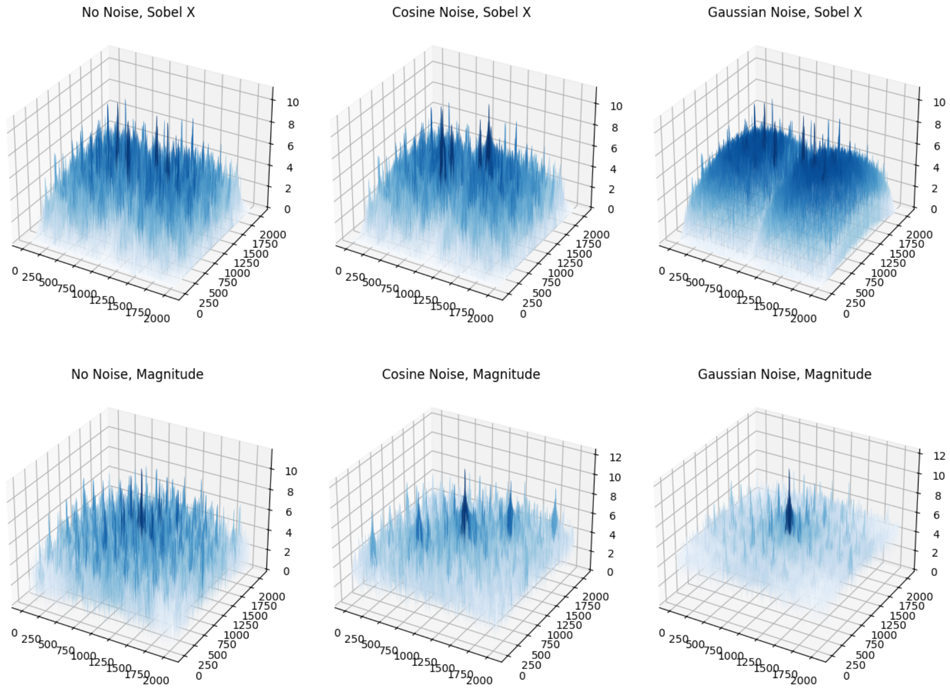

Figure 3: Frequency Response of the Sobel-edge-detected, rotated checkerboard with varying noise-types

The Laplacian operator produces sharp but fragmented edges due to unstable zero crossings. This leads to unusable results when noise is introduced. On the other hand Marr-Hildreth method improves robustness through prior smoothing, but achieves worse results in the checkerboard test-case without noise.

The Canny detector yields the most consistent results, producing thin and continuous edges while suppressing noise-induced responses. Non-maximum suppression reduces edge thickness, and hysteresis thresholding preserves only structurally connected edges.

In Figure 4 the real-world images were edge-detected by all of the above mentioned detectors. As expected the Canny edge-detector performed overall the best, while the instability of the Laplacian operator with zero crossing did not result in any meaningful edges.

Altogether the experimental observations lead to the conclusion, that derivative operators provide simple edge detection with limited robustness, while multi-stage

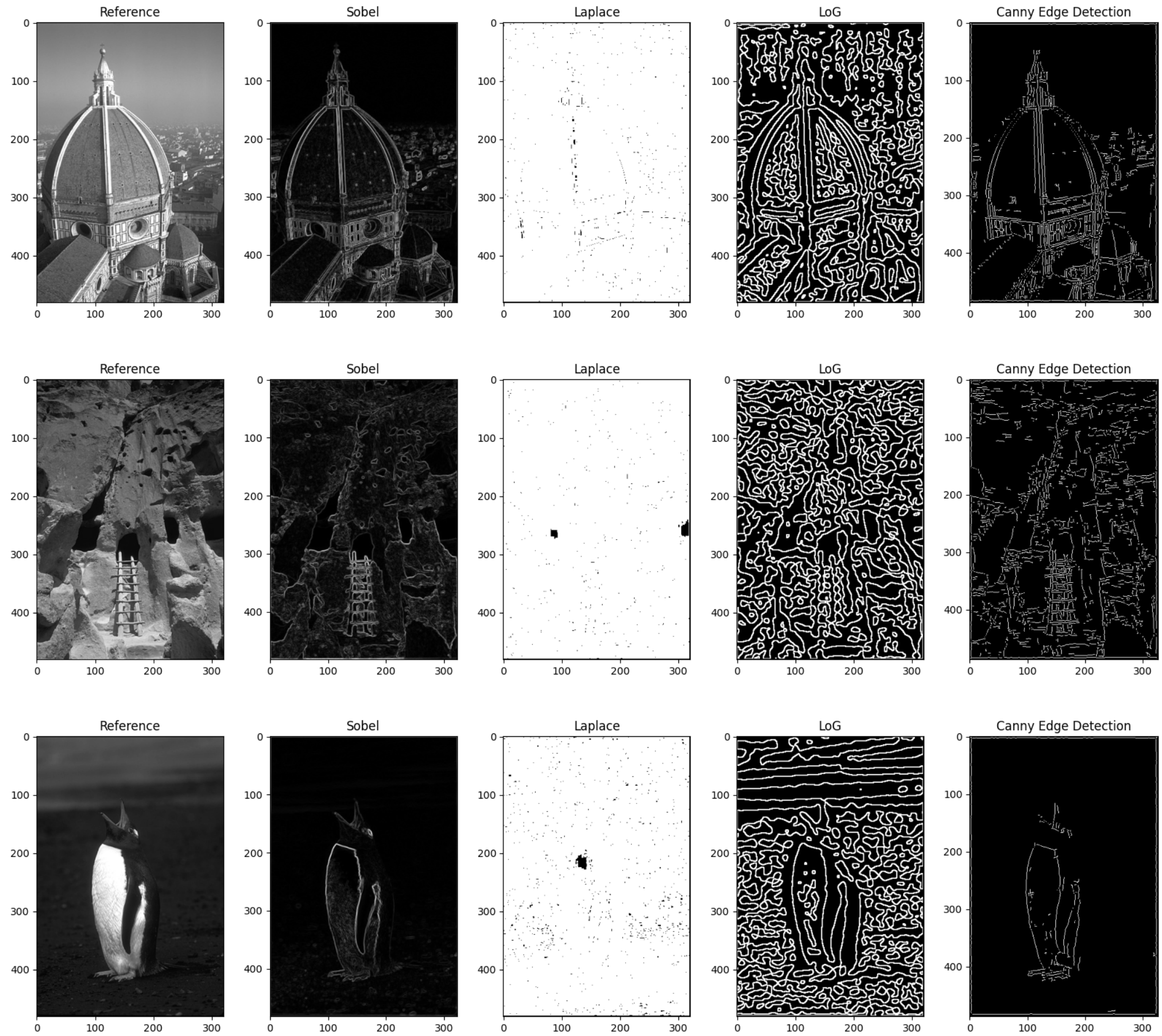

Figure 4: Edge-detection performed on real-world test images

approaches improve edge quality in real-word images.

Bibliography

[1] Thomas B. Moeslund. *Canny Edge Detection*. 2009.